

Module Document: Advanced Critical Care Nursing Science II

Institution: Ngee Ann Polytechnic – School of Health Sciences

Module Overview

Course Name:	ADVANCED DIPLOMA IN CRITICAL CARE NURSING – SPECIALTY MODULE 3
Module Name:	Advanced Critical Care Nursing Science II
Module Catalogue and ID:	303ACNSII / 012148
Pre-Requisite:	Not applicable
Timetabled Hours:	60 hours
Contact Hours:	60 hours
Credit Units:	4
Assessment Weightage:	Continuous Assessment 100%

1. Synopsis and Aims

This module aims to provide students with an understanding and comprehensive foundation on the principles of critical care nursing practice and a continuity of basic nursing science from general nursing training. Gastro-intestinal, hepato-biliary, endocrine and nephrology diseases and altered conditions of these major body systems are covered in depth with integration of pathophysiology and applied human anatomy. Interpretation of relevant laboratory and diagnostic studies results are included and advanced nursing skills that are performed commonly in the acute care setting, this module serves to equip critical care practitioners with the necessary knowledge of diseases and skills for critical thinking to make informed clinical decisions.

2. Learning Outcomes

On completion of the module, the students should be able to:

- **2.1** Discuss the nursing care management* of critically ill patients with gastrointestinal and hepato-biliary disorders:
 - acute abdomen
 - abdominal trauma
 - gastrointestinal bleeding
 - cholecystitis and cholelithiasis
 - acute pancreatitis
 - hepatic failure

\$\$\$Note \^* - includes definition, pathophysiology, clinical manifestations, diagnostic tests, treatment (medical, surgical), complications, and nursing care in critical care\$\$\$
- **2.2** Discuss the nursing care management* of critically ill patients with endocrinal disorders:
 - diabetic ketoacidosis
 - hyperosmolar hyperglycemic non-ketotic syndrome
 - thyroid storm
 - myxedema coma
 - diabetes insipidus and syndrome of inappropriate anti-diuretic hormone
 - adrenal insufficiency
 - Cushing's syndrome

\$\$\$Note \^* - includes definition, pathophysiology, clinical manifestations, diagnostic tests, treatment (medical, surgical), complications, and nursing care in critical care\$\$\$
- **2.3** Discuss the nursing care management* of critically ill patients with thermoregulation disorders:
 - hypothermia/hyperthermia

\$\$\$Note \^* - includes definition, pathophysiology, clinical manifestations, diagnostic tests, treatment (medical, surgical), complications, and nursing care in critical care\$\$\$
- **2.4** Discuss the nursing care management* of critically ill patients with renal disorders:
 - acute kidney injury (AKI) / acute renal failure
 - chronic kidney injury (CKI) / chronic renal failure

\$\$\$Note \^* - includes definition, pathophysiology, clinical manifestations, diagnostic tests,

- treatment (medical, surgical), complications, and nursing care in critical care\$\$
- **2.5** Discuss the nursing care management* of the critically ill patients with hematological disorders:
 - oncologic emergencies
 - Superior vena cava syndrome
 - Hypercalcemia
 - Tumor lysis syndrome
 - Spinal cord compression
 - systemic inflammatory response syndrome
 - sepsis
 - septic shock
 - disseminated intravascular coagulation

\$\$Note \^* - includes definition, pathophysiology, clinical manifestations, diagnostic tests, treatment (medical, surgical), complications, and nursing care in critical care\$\$
- **2.6** Discuss the nursing care management* of critically ill patients with drug overdose and poisoning.

\$\$Note \^* - includes definition, pathophysiology, clinical manifestations, diagnostic tests, treatment (medical, surgical), complications, and nursing care in critical care\$\$
- **2.7** Outline the nursing care management* of the patient undergoing gastrointestinal monitoring, diagnostic test, or procedure:
 - Sengstaken-Blakemore tube in esophageal varices;
 - Endoscopic procedures;
 - Gastrointestinal surgeries (includes ostomies/stomas)

\$\$Note \^* - where applicable, to include interpretation of waveforms/findings, identifying problems, troubleshooting, and nursing care – before, during, and after the procedure.\$\$
- **2.8** Outline the nursing care management* of the patient undergoing renal procedures:
 - renal replacement therapies;
 - peritoneal dialysis;
 - hemodialysis;
 - continuous kidney replacement therapy;

\$\$Note \^* - where applicable, to include interpretation of waveforms/findings, identifying problems, troubleshooting, and nursing care – before, during, and after the procedure.\$\$
- **2.9** Apply the principles of dialysis with the following renal replacement therapy (RRT):
 - (a) continuous kidney replacement therapy (CKRT);
 - (b) intermittent hemodialysis; and
 - (c) peritoneal dialysis

3. Content: Topics and Hours

Online e-learning hours*: 30 hours

No	Topic	Lecture-Tutorial hours
1	GASTROINTESTINAL – Acute abdomen (includes intestinal obstruction and peritonitis) – Abdominal trauma and compartment syndrome – Gastrointestinal bleeding	3 3*
2	GASTROINTESTINAL – Cholecystitis and cholelithiasis – Acute pancreatitis – Hepatic failure	3 3*
3	GASTROINTESTINAL – Endoscopic procedures – Gastrointestinal surgery (includes ostomies/stomas) – Sengstaken-Blakemore tube in esophageal varices	3 3*
4	ENDOCRINE Diabetic disorders – Diabetic ketoacidosis	4 4*

	(DKA) – Hyperosmolar hyperglycemic non-ketotic syndrome (HHNS) Thyroid disorders – Thyroid storm/thyroid crisis – Myxedema coma Pituitary disorders – Diabetes insipidus and Syndrome of Inappropriate Anti-Diuretic Hormone (SIADH) Adrenal disorders – Adrenal insufficiency – Cushing's syndrome	
5	THERMOREGULATION – Hypothermia – Hyperthermia	3 3*
6	RENAL – Acute kidney injury (AKI) – Chronic kidney disease	4 4*

	(CKD)	
7	RENAL – Principles of renal replacement therapy (RRT) – Renal replacement therapy: * Continuous kidney replacement therapy (CKRT) * continuous venovenous hemofiltration (CVVH); * continuous venovenous hemodialysis (CVVHD); and * continuous venovenous hemodiafiltration (CVVHDF) * intermittent hemodialysis (HD) * peritoneal dialysis (PD)	4 4*
8	HEMATOLOGY – Oncologic emergencies * Superior vena cava syndrome * Hypercalcemia * Tumor lysis syndrome * Spinal cord compression	4 4*

	– Systemic inflammatory response syndrome (SIRS) – Sepsis – Septic shock – Disseminated intravascular coagulation (DIC)	
9	DRUG OVERDOSE AND POISONING	2 2*
	Overall total	60

4. Teaching Methodology / Teaching & Learning Strategies

Case-Based Learning

In Case-Based Learning, students are typically posed a realistic story or set of circumstances, or complex situation. Students will actively learn to integrate knowledge, skills and attitudes using analytical and critical thinking skills to make decisions or solve issues at various points. They are encouraged to explore assumptions and alternatives, or even future factors, to arrive at different solutions or conclusions just as they would do on the job, except that this learning is set in a safe but real-world context. Numerous feedback opportunities to students are provided, based on the decisions they make at each stage in the process.

ICT-enabled learning

The module uses a wide range of synchronous and asynchronous RL tools to facilitate student-to-student, teacher-to-students or student-to-content interaction, online. The various forms of RL could include recorded lectures, interactive PowerPoints, videos, online resources, online quizzes, online discussion boards, POLITEMALL, Zoom, and other interactive courseware mounted on the learning management system. The learning activities are tracked in the learning management system.

5. Assessment Strategies

This is a continuous assessment module and the weightages for the assessments are as the following:

- **Group Project: 50%**

The students are required to explore and critically analyse a case scenario. A variety of approaches and delivery methods can be used for the group project presentation.

Demonstration of reflective learning, clinical reasoning and synthesis of concepts learnt in the module are expected. The assessment encompasses **30% group and 20% individual components.**

- **Class Test 1: 25%**

- **Class Test 2: 25%**

For class tests, topics to be tested will be announced to students. Students will be required to revise and read widely to be prepared for the test.

6. Plagiarism Policy

The Ngee Ann Polytechnic Plagiarism policy mandates that students must acknowledge the original author when they submit work. Failing to do so will constitute plagiarism.

Students are expected to reference their individual case study report and case presentations slides as per American Psychology Association (APA) format. Students assume full responsibility for the content and integrity of the academic work they submit. A student's submitted work must be that of his/her work. Students shall be guilty of violation if they:

- (a) Represent the work of others as their own
- (b) Use or obtain unauthorized assistance in any academic work
- (c) Give unauthorized assistance to other students
- (d) Misrepresent the content of submitted work

Students who commit the plagiarism offence are subject to the disciplinary procedures outlined in the Ngee Ann Plagiarism Policy.

7. Recommended Texts / References / Learning Resources

The reference list here is carefully selected but by no means a complete list. Students are encouraged to read widely and to share good articles with the group. Recommended articles for tutorials will be included in the tutorial package.

7.1 Textbooks

- Aitken, L., Marshall, A., & Chaboyer, W. (2019). *ACCCN's Critical Care Nursing*. (4th ed.). Chatswood, N.S.W.: Elsevier Australia.
- Baird, MS. (2021). *Manual of Critical Care Nursing: Interprofessional Collaborative Management*. (8th ed.). St. Louis, Mo.: /Mosby.
- Diepenbrock, N. H. (2020). *Quick Reference to Critical Care*. (6th ed.). Philadelphia, PA: Lippincott /Williams /Wilkins.
- Johnson, K. L (2023). *AACN Procedure Manual for Progressive and Critical Care*. (8th ed.).

Elsevier.

- Morton, P. G., & Thurman, P. (2023). *Critical Care Nursing: a holistic approach*. (12th ed.). Philadelphia: Wolters Kluwer Health/Lippincott Williams & Wilkins.
- Sole, M. L., Klein, D., & Moseley M. (2020). *Introduction to Critical Care Nursing*. (8th ed.). St. Louis, Mo.: Saunders/Elsevier.
- Thomas, N. (2019). *Renal nursing*. (5 ed.). Chichester, West Sussex: Wiley Blackwell.
- Urden, L. P., Stacy, M. K., & Lough, M. E. (2021). *Critical Care Nursing: diagnosis and management*. (9th ed.). St. Louis, Mo.: Saunders / Elsevier.

7.2 Journals

- American Association of Critical Care Nurse (AACN) Advanced Critical Care
- American Journal of Critical Care
- American Journal of Respiratory and Critical Care Medicine
- Critical Care Nursing Journal
- Journal of Advanced Nursing
- Journal of Critical Care
- Journal of Trauma Nursing